

AI & ML Internship Program

Task 13: Digital Payment Analytics

Domain: Data Analytics + FinTech Analytics + Business Intelligence

Difficulty: Intermediate

Industry Focus: FinTech / Banking / Digital Payments / Customer Intelligence

Skills Covered: KPI Design, Dashboard Development, Transaction Analytics, User Segmentation, Trend Analysis, Data Storytelling

Business Problem

Digital payment platforms process millions of transactions every day through:

- UPI
- Wallet payments
- QR payments
- Credit/Debit cards
- Bank transfers

As transaction volume grows, payment companies face major business challenges:

- Failed transactions
- Fraud risks
- Customer drop-offs
- Peak-hour load issues
- User retention problems

Leadership teams need answers:

- Which payment methods are used most?
- When do transaction failures increase?
- Which customer segments generate highest revenue?
- What transaction patterns indicate operational problems?
- Which regions require attention?

As a Data Analyst, your responsibility is to build an analytics system that helps teams monitor digital payment performance.

Real-world Scenario

You are working as a Data Analyst for a digital payment company.

Recently, customer complaints related to payment failures and transaction delays increased by **22%**.

Leadership wants an analytics dashboard to monitor payment operations, identify risks, and improve customer experience.

Your mission is to investigate transaction patterns and build actionable insights.

Objective

Build a Digital Payment Analytics Dashboard to analyze transaction behavior, customer activity, payment trends, and operational KPIs.

The dashboard should support business and risk teams.

Dataset Suggestions

Choose one:

- Digital payment datasets
- UPI transaction datasets
- FinTech transaction datasets
- Banking datasets
- Generate synthetic payment data

Tasks to Perform

Phase 1: Data Understanding & Cleaning

- Load dataset
- Handle missing values
- Remove duplicates
- Convert date/time columns
- Create derived columns

Examples:

- Peak Hour Flag
- High-Value Transaction Flag

- Daily Usage Score
- Failure Rate Indicator

Phase 2: KPI Design

Create important business KPIs:

Mandatory KPIs:

- Total Transactions
- Total Payment Volume
- Success Rate
- Failure Rate
- Average Transaction Value
- Peak Transaction Hour
- Active Users
- Revenue Contribution by Payment Method

Questions:

- Which KPIs define payment health?
- Which KPIs help operations teams?

Phase 3: Analytics Investigation

Analyze:

- Payment Method vs Volume
- City vs Transaction Count
- Time vs Failure Rate
- Device Type vs Usage
- Merchant Category vs Revenue

Visualizations:

- KPI cards
- Heatmaps
- Bar charts
- Line charts
- Funnel analysis

Questions:

- Which payment methods dominate?
- When do failures spike?
- Which categories generate highest volume?

Phase 4: User Segmentation

Create customer categories:

Examples:

- Heavy users
- High-value users
- Frequent transactors
- Cashback-driven users
- Low-engagement users

Analyze behavioral differences.

Phase 5: Dashboard Development

Create sections: **Executive Overview**

Include:

- Total volume
- Total users
- Revenue metrics
- Success rate

Transaction Dashboard

Analyze:

- Payment trends
- Transaction volume
- Time-based activity

Customer Analytics Dashboard

Analyze:

- User segments
- Customer spending patterns

- Active user metrics

Risk Dashboard

Analyze:

- Failed transactions
- Failure spikes
- High-value transactions

Business Questions to Answer

1. Which payment methods perform best?
2. Which time periods experience failures?
3. Which users contribute most revenue?
4. Which regions require operational improvement?
5. What actions should product and operations teams take?

Tools Allowed

Choose one or multiple:

- Power BI
- Excel
- SQL
- Python
- Tableau
- Streamlit

Expected Deliverables

- ☒ Interactive dashboard
- ☒ Analysis notebook
- ☒ KPI report
- ☒ Business recommendations
- ☒ GitHub repository
- ☒ README documentation

GitHub Submission Structure

TASK_05_Digital_Payment_Analytics/

- | — data/
- | — notebook/
- | — dashboard/
- | — images/
- | — reports/
- | — README.md
- | — requirements.txt

README should include:

1. Problem Statement
2. Dataset Description
3. KPI Definitions
4. Dashboard Screenshots
5. Key Findings
6. Recommendations
7. Future Scope

Evaluation Criteria

Criteria	Weight
Business Understanding	20%
Dashboard Design	20%
KPI Selection	20%
Insights	20%
Documentation	10%
Innovation	10%